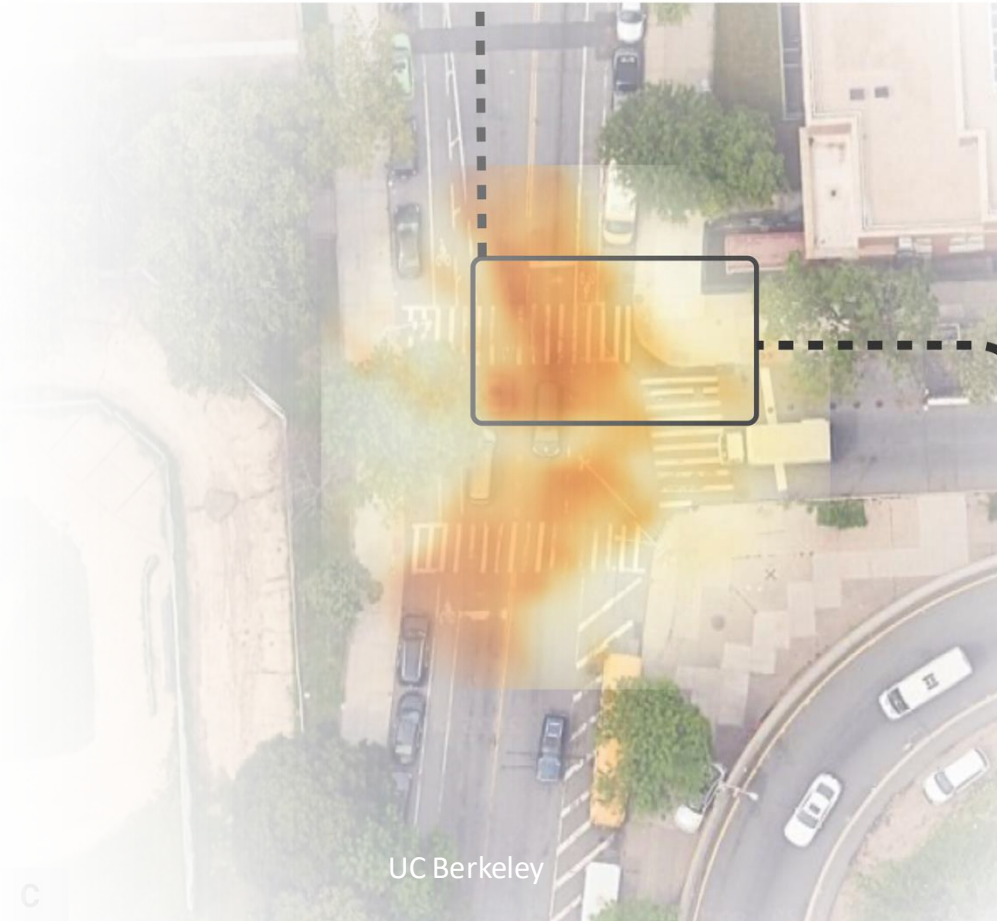
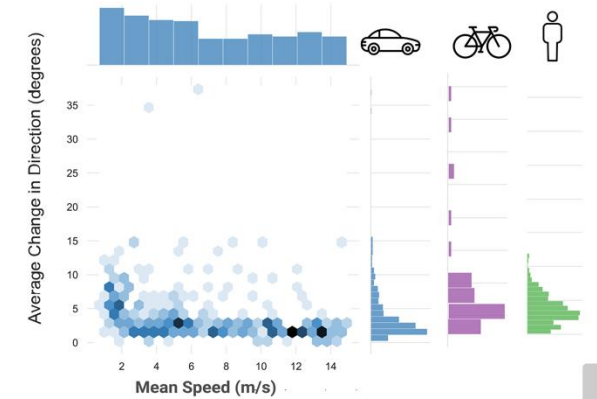
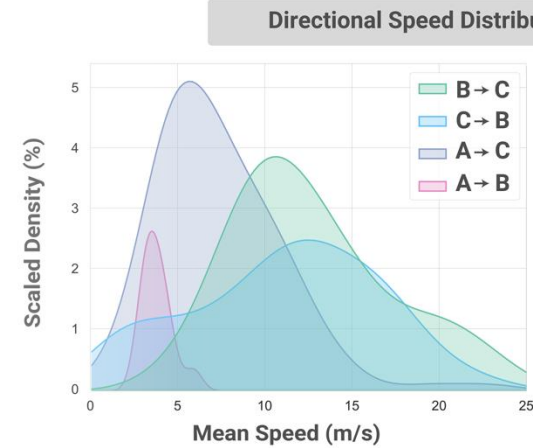
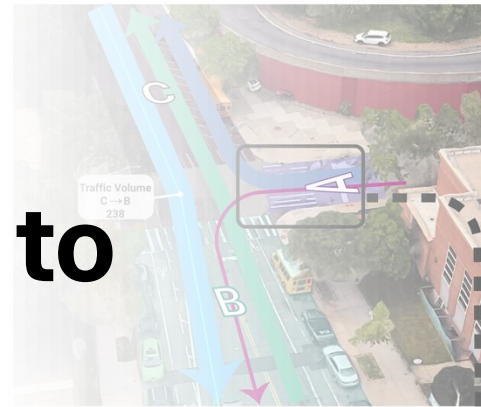


Introduction to Urban Data Analytics

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Data Exploration



Exploratory Data Analysis (EDA)

Exploratory data analysis (EDA) is an open ended, inductive approach used to examine data, identify patterns, detect anomalies, assess assumptions, and generate questions or hypotheses before formal testing. It focuses on discovering what the data may be showing rather than confirming a prespecified claim.

- What patterns, relationships, anomalies, or gaps appear in the data?
- What might these patterns mean, and what alternative explanations could produce them?
- Exploration **without attention to measurement, representation, and data generation** can reveal a real pattern in the dataset that *does not accurately describe the city*.

Confirmatory Data Analysis (CDA)

Confirmatory Data Analysis (CDA) is a targeted, hypothesis-driven statistical methodology used to test specific assumptions or theories about a population based on sample data. It serves as the rigorous, deductive phase of data analysis that follows exploratory data analysis (EDA), *focusing on validating patterns rather than discovering new ones.*

- Does a prespecified claim survive formal testing?
- Does the data and analysis support the claim I am making about the city?
- Confirmation **without valid measurement** *can confidently confirm the wrong thing.*

Fundamental Strategies for Creating Datasets

1. Collecting data on *everyone* and *everything* for the objects you are interested in
 - Examples: Census, parcel inventory, street network, a shop's inventory
2. Collecting data on just *some* of the items in population
 - Examples: Pedestrian counts, household travel surveys, air quality monitoring
3. Changing conditions: Deliberately change a condition and observe how outcomes respond, allowing comparison with what would have happened otherwise.
 - Examples: Street-design experiment, parking pricing, Change fertilizer and measure crop growth.

Observational

Experimental

Coverage Does Not Guarantee Completeness

Collecting everything does not mean observing everything that matters.

- Some records or values may still be missing.
- Some relevant characteristics may never be measured.
- Definitions determine what counts and what does not.
- Measurement itself can introduce error and uncertainty.

Example: A complete street database can still omit informal paths, inaccessible crossings, or current sidewalk conditions.

Key question:

What about the city is still invisible in this dataset?

What Data Measures?

Observed data are produced through processes of inclusion and exclusion.

- **Selection:** Which people, places, or times were chosen?
- **Self selection:** Who chooses to participate or report?
- **Nonresponse:** Who was approached but is missing?
- **Coverage:** Who never had a chance to appear?
- **Examples:** Pedestrian counts at selected intersections $\neq$ all pedestrian activity
311 complaints $\neq$ all neighborhood problems
Travel survey respondents $\neq$ everyone who travels

- **Key question:**

*Who is missing, and **could their absence systematically change what we see?***

The Measure is Not the Concept

Data usually measures an operationalization or proxy, not the phenomenon itself.

We observe	We want to know
Crash counts	Traffic safety / risk
311 complaints	Infrastructure need
Transit stops	Accessibility
Recorded trips	Mobility
Building permits	Development activity

- **A precise measure of the wrong thing is still the wrong measure.**
- **Key Question:** *What does one observation actually represent?*

What Follows From the Data?

Patterns in observed data do not automatically generalize to the population or phenomenon of interest.

EDA: What structure, variation, anomalies, or relationships appear in the observations?

CDA: Does a prespecified hypothesis withstand formal statistical testing?

Responsible analysis: Are the data, measurements, design, and assumptions sufficient to support the claim being made?

- Observed pattern $\neq$ population pattern
- Correlation $\neq$ Causal effect
- Statistical evidence $\neq$ Measurement validity

The strength of a conclusion cannot exceed the strength of the data and design supporting it.

Sanity Checks in Pandas



Raw Data is Read-only



You should *always* make a copy of your dataset before doing any data preparation or cleaning in case a change you make is incorrect and you need to refer to the original.



Sanity Checks: Initial Setup Checks

1) Quick structure scan

- Preview rows and dtypes: `df.head()`, `df.tail()`, `df.info()`
- Random spot-check: `df.sample()`

2) Column names: `df.columns`

- Remove unwanted characters (e.g., `#` `%` `@`) and whitespace
- Standardize case (lowercase is simplest)
- Rename systematically if needed
- Example: `df.columns = (df.columns.str.strip() # trim spaces
.str.lower() # lowercase .str.replace(r'^0-9a-z_', '_', regex=True)) # safe chars`



Sanity Checks: Initial Setup Checks

3) Index hygiene : `df.index`

- Ensure uniqueness: `df.index.is_unique`
- Check type for consistency (string / numeric / timestamp)
- Reset if necessary: `df = df.reset_index(drop=True)`
- Or set a meaningful index: `df = df.set_index("column_name")`



Sanity Checks: Data Quality Checks

Identify and fix common data issues to ensure accuracy.

1) Duplicates

Count: `df.duplicated().sum()`

Remove: `df.drop_duplicates()` # Can I remove these?

2) Missing Values: `df.isnull().sum()`

3) Invalid values

Look for impossible or inconsistent entries (e.g., negative prices, incorrect dates)

4) String formatting: Standardize capitalization, trim whitespace, etc.

5) Data types: Ensure numbers are stored as numeric types (`df.dtypes`).



Sanity Checks: Data Distribution & Summary Statistics

Check central tendencies and spread

1) Descriptive Statistics

`df.describe()` for count, mean, std, min, max, and quartiles.
Useful overview but beware of relying solely on these measures.

2) Histograms

`df.hist()` to visualize distributions
Identify skewness, gaps, or uniformity.



Sanity Checks: Outlier & Anomaly Detection

Catch unusual values that may distort results.

1) Box Plots

```
sns.boxplot(x=df["column"]) # Quick visual cue for outliers
```

2) IQR Method

Outliers fall outside: $[Q1 - 1.5 \times IQR, Q3 + 1.5 \times IQR]$

```
Q1 = df['value'].quantile(0.25)
```

```
Q3 = df['value'].quantile(0.75)
```

```
IQR = Q3 - Q1
```

3) Domain Knowledge

Validate edge cases (e.g., $\text{len}(\text{FIPS}) \leq 15$?)

Let's Practice!



Open W06 Lecture Notebook